

MEDICAL INSURANCE COST

Statistical Analysis & Interactive Dashboard

Lab-4: Applied Statistical Modeling & Interactive Web Dashboard

Course: Statistical Modeling with Python

Course Code: DS602

Program: M.Sc. Data Science

University: Dhirubhai Ambani University

Semester: 1

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GitHub Repository: [View on GitHub](#)

Live Application (Streamlit): [Open Dashboard](#)

1. Project Overview

This project is an end-to-end applied statistical modeling workflow that transitions static data analysis into a production-ready application. It bridges theoretical statistics and software engineering by converting exploratory data analysis (EDA) and inferential testing from a Jupyter Notebook into a fully interactive, decoupled Streamlit web dashboard.

2. Project Objective

To architect a dynamic data tool that evaluates the financial drivers of medical billing. The application automates parametric and non-parametric hypothesis testing, fits an Ordinary Least Squares (OLS) regression model, and generates real-time patient cost predictions alongside rigorous Gauss-Markov residual diagnostics.

3. Dataset

The analysis utilizes the insurance.csv dataset, which contains 1,338 distinct patient records. It consists of 7 attributes designed to evaluate how demographics and lifestyle choices impact healthcare costs.

Category	Attribute(s)	Type
Target Variable	charges	Continuous
Demographics	age	Continuous
Demographics	sex, region	Categorical
Health Metrics	bmi	Continuous
Health Metrics	smoker	Categorical
Dependents	children	Discrete

4. Technologies and Libraries

Layer	Technology / Library
Language	Python 3
Frontend UI	Streamlit
Statistical Modeling & Testing	Statsmodels, SciPy
Data Manipulation	Pandas, NumPy
Data Visualization	Plotly (interactive), Matplotlib (static diagnostics)

5. Project Structure and Environment

The project is built for a standard virtual environment (venv) deployment and managed via Git. Verified against the live repository — note the notebook file name below differs from earlier drafts:

- **app.py** — The core Streamlit application housing the dashboard UI and live inference logic.
- **code.ipynb** — The Jupyter Notebook used for initial data profiling, statistical assumption testing, and model prototyping. (Corrected: this file is named code.ipynb, not code_2.ipynb.)
- **data/insurance.csv** — The raw dataset directory.
- **requirements.txt** — The dependency file required for deployment to Streamlit Community Cloud.

6. Exploratory Data Analysis (EDA)

The EDA module provides real-time data filtering based on age, region, and smoking status. It dynamically recalculates measures of central tendency (mean, median), dispersion (standard deviation, IQR), and shape (skewness). Visual diagnostics include Plotly-powered histograms with marginal box plots to assess feature distributions, and a bivariate correlation heatmap to identify preliminary linear relationships.

7. Hypothesis Testing

An automated testing engine evaluates group differences at a strict alpha level of $\alpha = 0.05$. The pipeline rigorously checks mathematical assumptions before executing inference.

- **Shapiro-Wilk Test (Normality Validation):** Evaluates whether the continuous target variable (e.g., medical charges) within a specific group follows a normal, Gaussian distribution.
- **Levene's Test (Homoscedasticity):** Assesses whether the variance in the target variable is statistically equal across the groups being compared.
- **Two-Sample t-test (Parametric):** The standard protocol for comparing two independent groups, applied strictly *only* if the data passes both the Shapiro-Wilk and Levene's tests.
- **Mann-Whitney U Test (Non-Parametric Fallback):** Automatically triggered if the data violates normality or equal variance assumptions. In this analysis, medical charges were heavily skewed; this test confirmed a massive, statistically significant difference in costs between smokers and non-smokers ($p < 0.001$).
- **One-Way ANOVA (Multi-Group):** Utilized to analyze variance across categorical factors with three or more distinct levels. This test confirmed significant geographic pricing variation across the four U.S. regions ($p = 0.0309$).

8. Multiple Linear Regression

An Ordinary Least Squares (OLS) regression model dynamically fits the formula $\text{charges} \sim \text{age} + \text{bmi} + \text{children} + \text{C}(\text{sex}) + \text{C}(\text{smoker}) + \text{C}(\text{region})$ to predict medical costs based on user inputs.

- **Model Performance:** The regression yields an R^2 of 0.751, demonstrating strong explanatory power by capturing 75.1% of the variance in medical charges.

- **Primary Coefficient (Smoker):** Smoking is the single largest financial driver. Holding all other attributes constant, identifying as a smoker increases expected baseline medical charges by approximately \$23,850.
- **Continuous Coefficients:** Age and BMI demonstrate statistically significant positive correlations ($p < 0.001$). Every additional year of age adds \$256.86 to the predicted cost, and every one-unit increase in BMI adds \$339.19.
- **Live Inference:** The application generates real-time predicted medical charges from user slider inputs, outputting mathematically rigorous 95% Confidence and Prediction Intervals.
- **Gauss-Markov Diagnostics:**
 - **Variance Inflation Factor (VIF):** Analysis confirms zero multicollinearity among the continuous predictors (all VIF scores ≈ 1.0).
 - **Jarque-Bera Test:** Assesses residual normality. The test returns a p-value < 0.001 , flagging non-normal errors and suggesting the potential need for future log-transformations.
 - **Visual Diagnostics:** Verifies homoscedasticity and linearity via live-rendered Residuals-vs-Fitted plots and Normal Q-Q plots.

9. Links

GitHub Repository: https://github.com/Khushal0911/Khushal-Bafna_202618005_DS602_Statistical-Methods

Live Streamlit Application: <https://202618005-ds602-statistical-methods-kb.streamlit.app/>